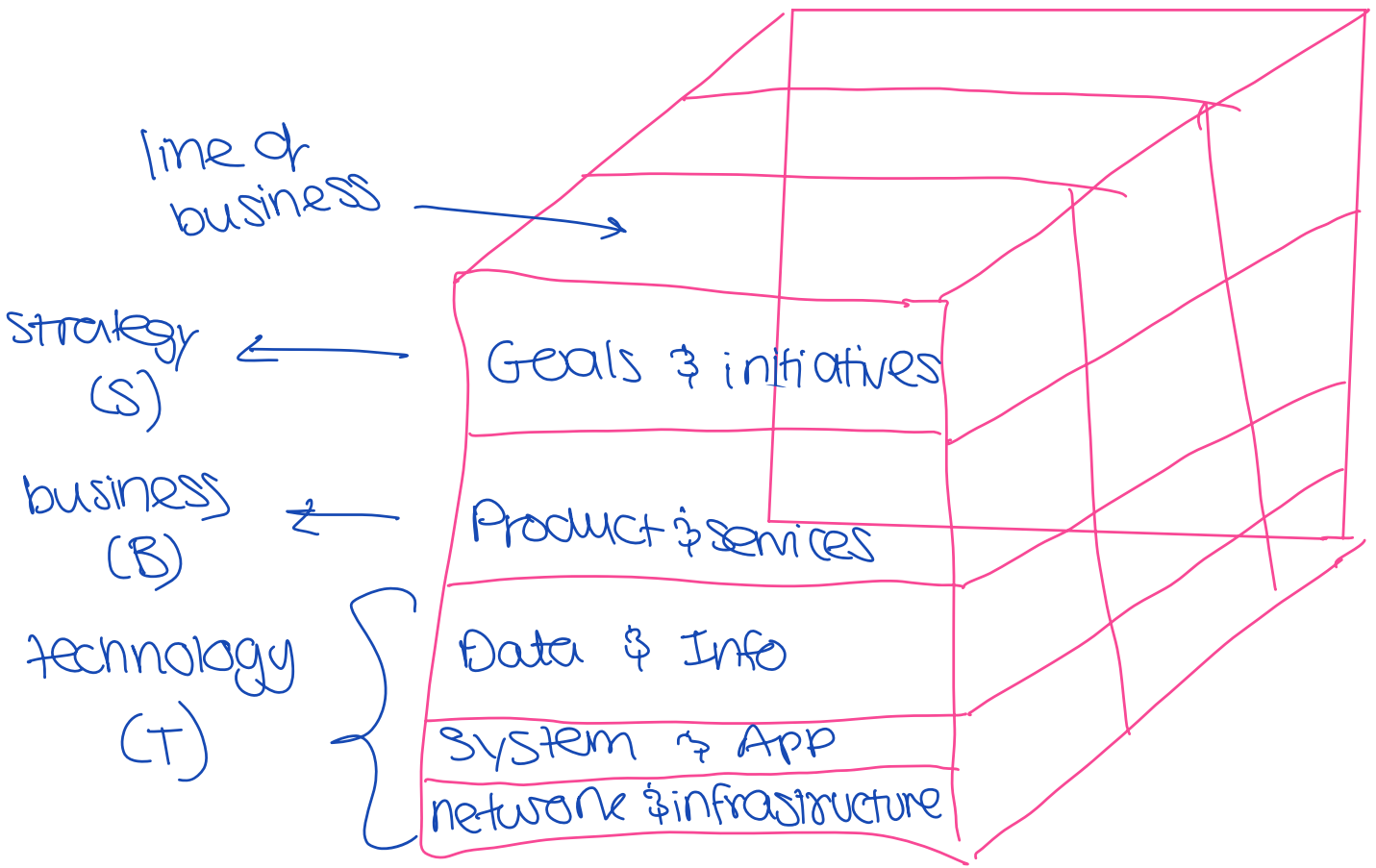
CHAPTER 01

Enterprise Architecture
As an Analysis & Design Method

EA 3 Framework



↳ all three levels of 3 S's should be engaged

→ dimension 1: multiple domains within a sub-arch falling under one line of business

→ Dimension 2: o on all artifacts each line

EA^3 Cube Framework

→ used to organize & document an entire enterprise so top managers and executives:

- understand the flow of the organization
- its connection to technology, strategy, business operations

x it is NOT merely an IT plan
→ all EA components work hand-in-hand collectively
technology is leveraged based on organizational goals and business requirements
and NOT arbitrarily investing in needless technology

Curses | of

EA as an Analysis and Design Method

not important

- References to EA began to emerge in the late **1980's** in management and academic, with focus on technical or systems architectures and schemas for organizing information.
- The concept of EA analysis and design emerged in the early **1990's** and has evolved to include views of strategic goals, business services, information flows, systems and applications, networks, and the supporting infrastructure.
- Additionally, there are **'threads'** that spread through every level of the architecture: standards, security, and skills.

EA as an Analysis and Design Method(cont)

EA analysis and design are accomplished through the following six basic elements:

(1) EA documentation framework

(2) Implementation methodology that support the creation

(3) Current views

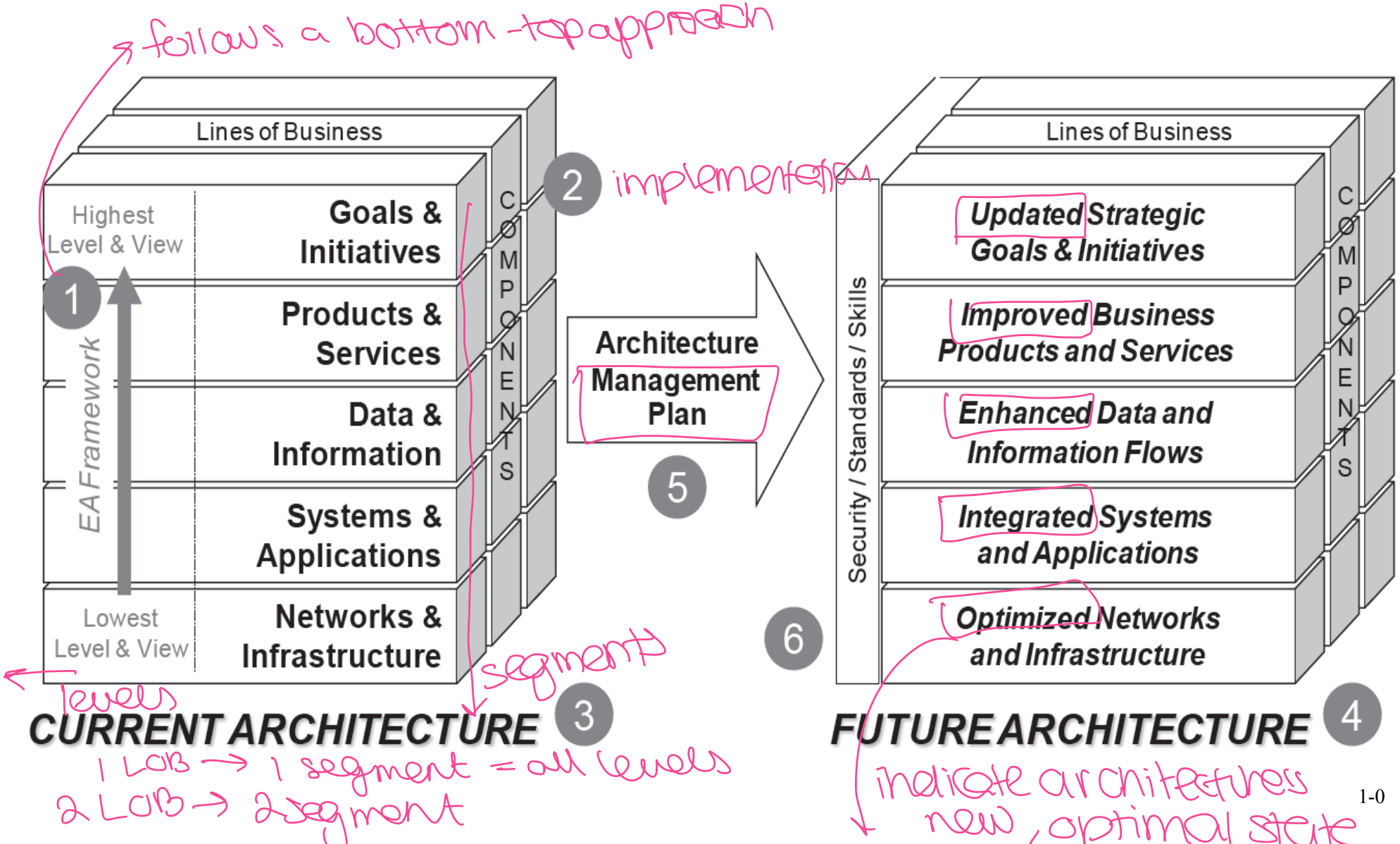
(4) Future views of the architecture

(5) The development of EA Management Plan to manage the enterprise's transition from ***current*** to ***future*** of the architectures.

There are also several areas common to all levels of the framework that are referred to as

(6) "threads" as shown in Figure (Next Page).

EA as an Analysis and Design Method(cont)



can be combined to form one segment

EA Analysis and Design Element #1:

The Framework.

1

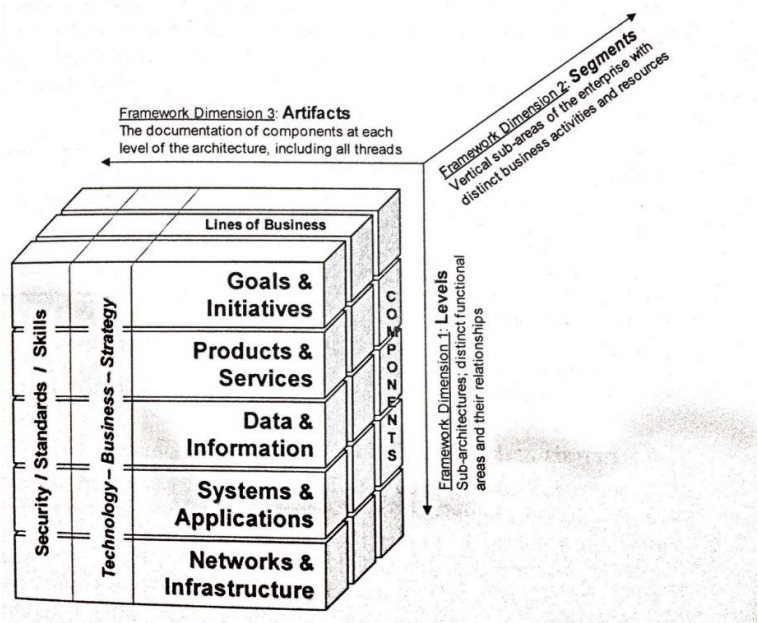
- The **EA framework** identifies the scope of the architecture to be developed and establishes relationships between the architecture's areas.
- The framework's scope is reflected through its geometric design and the areas that are identified for documentation.
- The framework creates an abstracted set of 'views" of an enterprise through the way that it collects and organizes architecture information.
- To lower risk and promote efficient, phased implementation methods, the EA framework is divided into segments of distinct activity, referred to as **Lines of Business** (LOBs).

EA Analysis and Design Element #1:
The Framework (cont)

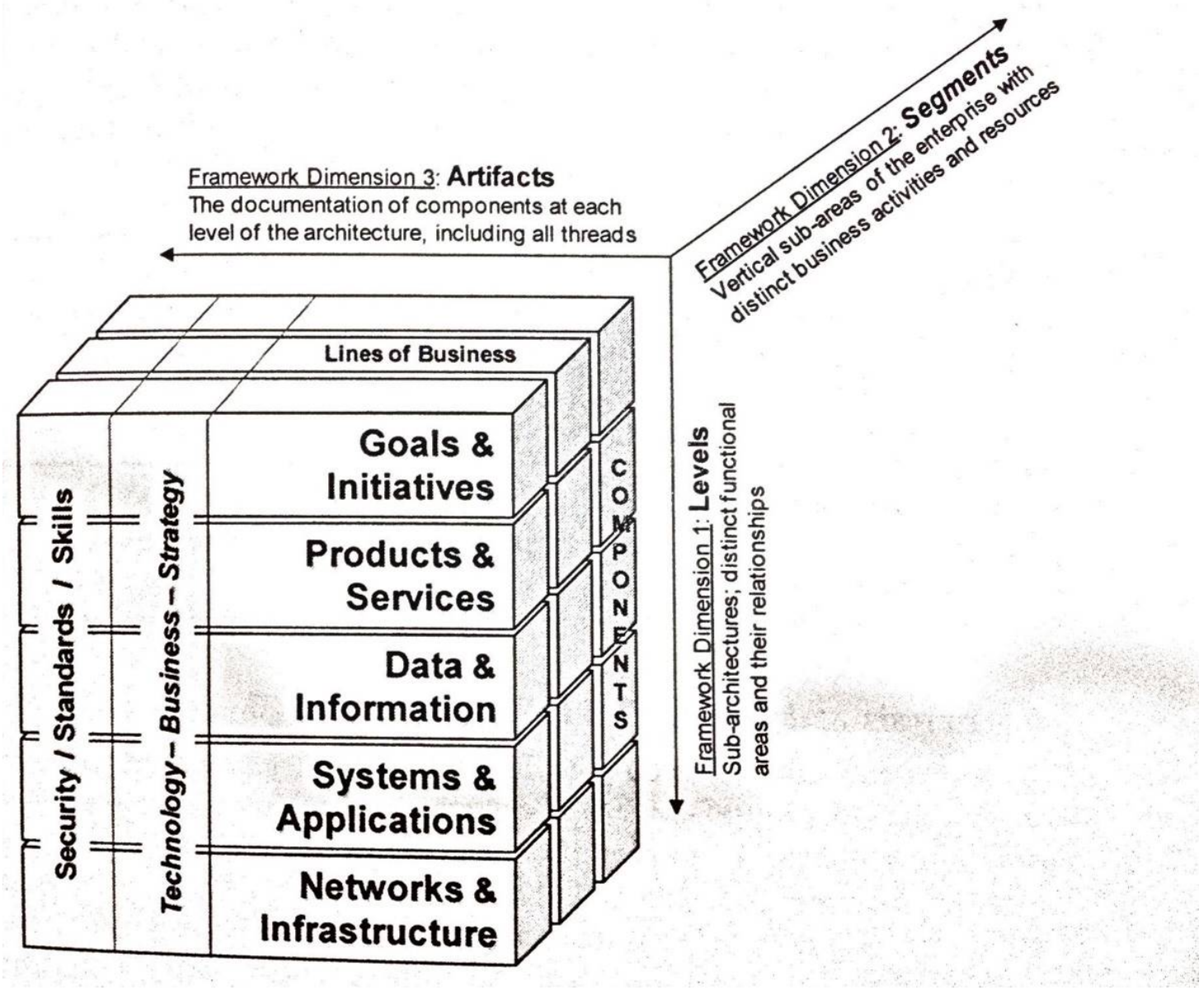
resembles waterfall methodology

- Known as the **EA³ Cube Framework™** the levels of this example framework are hierarchical so that the different sub-architectures (that describe distinct functional areas) can be logically related to each other.
- This is done by positioning high-level strategic goals/initiatives at the top, business products/services and data/information flows in the middle and supporting systems/applications and technology/infrastructure at the bottom. In this way alignment can be also be shown between strategy, information, and technology, which aid planning and decision-making.

A segment can exist as a stand-alone part of the EA.



EA3 Cube Analysis & Design Framework



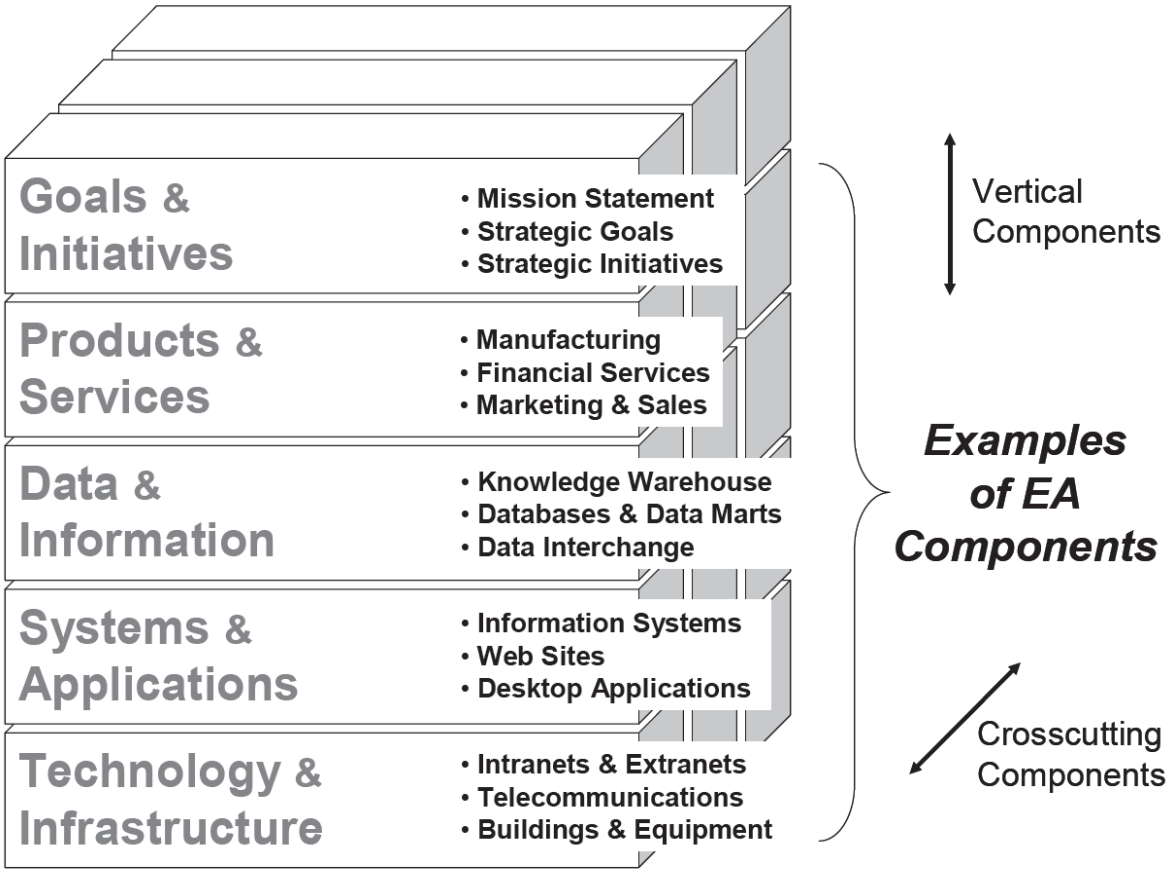
EA Analysis and Design Element #2:
EA Components

EA components are changeable goals, processes, standards, and resources that may extend enterprise-wide or be contained within a specific line of business or segment.

- **Examples** of components include strategic goals and initiatives; business products and services; information flows, knowledge warehouses, and data objects; information systems, software applications, enterprise resource programs, and web sites; voice, data, and video networks; and supporting infrastructure including buildings, server rooms, wiring runs/closets, and capital equipment.

EA Analysis and Design Element #2: EA Components (cont)

Examples of vertical and crosscutting (horizontal) EA components at each level of the EA³ Cube framework.



A **vertical component** is a changeable goal, process, program, or resource (equipment, systems, data, etc.) that serves one line of business.

A **horizontal (or crosscutting) component** is a changeable goal, process, program, or resource that serves several lines of business. Examples include email and administrative support systems that serve the whole enterprise.

EA Analysis and Design Element #3:
Current Architecture

- The current architecture or "**as-is**" view contains those **EA components** that **currently exist** within the enterprise at each level of the framework.
- The current view of the EA serves to create a 'baseline' inventory of current resources and activities that is documented in a consistent way with the future view of the EA so that analysts can see gaps in performance between future plans and the current capabilities.
- Having an accurate and comprehensive current view of **EA components** is an important reference for project planning, asset management, and investment decision-making.
- **The current view** of the EA is composed **of 'artifacts'** (documents, diagrams, data, spreadsheets, charts, etc.) at each level of the framework, which are archived in an **online EA repository** to make them useable by various EA stakeholders.

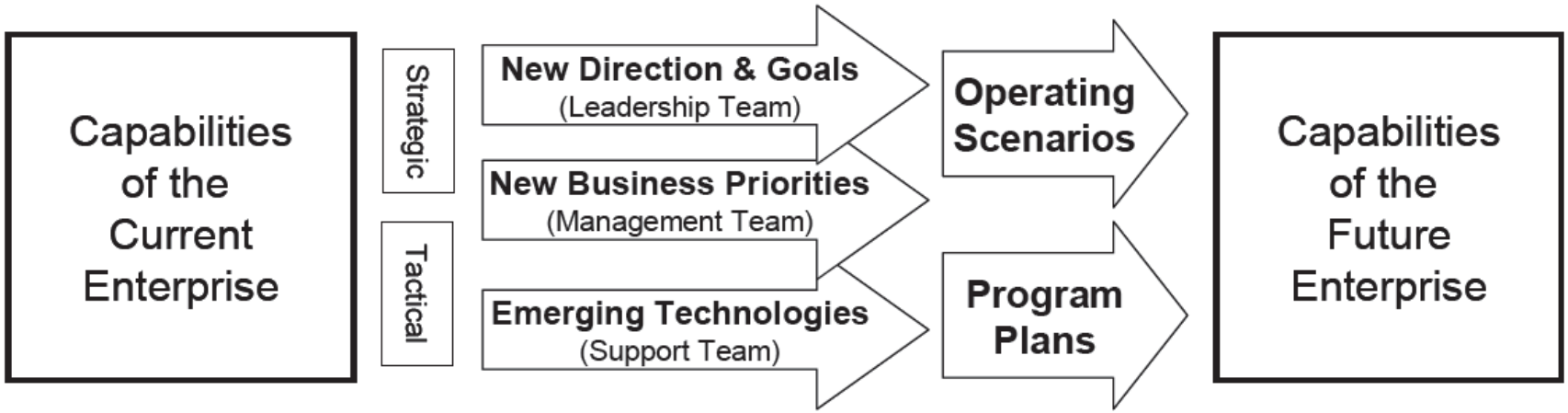
EA Analysis and Design Element #4:
Future Architecture

- The future architecture documents those **new or modified EA components** that are needed by the enterprise to close an existing performance gap or support a new strategic initiative, operational requirement, or technology solution.
- The future architecture should cover planned changes to EA components in the near term (tactical changes in the next 1-2 years), as well as changes to EA components that are a result of the implementation of long-term operating scenarios that look 3-10 years into the future (strategic changes).
- These scenarios **incorporate** different internal and external drivers and can help to identify needed changes in processes, resources, or technology that translate to future planning assumptions, which in turn drive the planning for new EA components.

EA Analysis and Design Element #4: Future Architecture (cont)

Future architecture is driven at both the strategic and tactical levels in **3 ways**: new directions & goals; changing business priorities; & emerging technologies.

The EA cannot reflect these changes in the future architecture:
- unless the enterprise's leadership team provides the changes in strategic direction and goals;
- unless the line of business managers and program managers provide the changes in business processes and priorities that are needed to accomplish the new goals; and
- unless the support/delivery staff identifies viable technology and staffing solutions to meet the new business requirements.



EA Analysis and Design Element #5:
EA Management Plan

- **The EA Management Plan** articulates the EA program and documentation approach.

top-down approach

bottom-up approach

- **The EA Management Plan** also provides descriptions of current and future views of the architecture, and a sequencing plan for managing the transition to the future business/technology operating environment.

migrate from current to future

- **The EA Management Plan** is a living document that is essential to realizing the benefits of the EA as a management program. How the enterprise is going to continually move from the current architecture to the future architecture is **a significant planning** and **management challenge**, especially if IT resources supporting key business functions are being replaced or upgraded.

EA Analysis and Design Element #6: Threads

- **EA documentation includes 'threads'** of common activity that are present in all levels of the framework. These threads include IT-related security, standards, and skill considerations.

- **Security:** Security **is most effective when it is an integral part of the EA management program and documentation methodology**. A comprehensive IT Security Program has several focal areas including: information, personnel, operations, and facilities. To be effective, IT security must work across all levels of the EA framework and within all of the EA components.

EA Analysis and Design Element #6: Threads (cont)

- **Standards:** One of the most important functions of the EA is that it provides technology-related standards at all levels of the EA framework. The EA should draw on accepted international, national, and industry standards in order to promote the use of non-proprietary solutions in EA components. This in turn enhances the integration of EA components, as well as better supporting the switch-out of components when needed.

- **Skills:** Perhaps the greatest resource that an enterprise has is **people**. It is therefore important to ensure that staffing, skill, and training requirements are identified for LOB and support service activities at each level of the EA framework, and appropriate solutions are reflected in the current and future architectures.

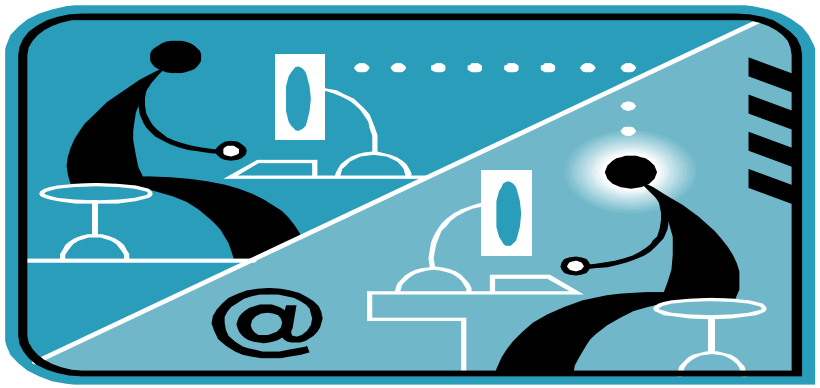
Key EA Concepts: Comparison & Clarification

Term	Definition	Scope in EA	Example
Level	A hierarchical layer of the EA³ Cube (from strategy to infrastructure).	Defines *what perspective* is being analyzed.	Goals, Business Services, Data, Systems, Infrastructure.
Line of Business (LOB)	Distinct functional area of the enterprise.	Column in the EA Cube.	Finance, HR, Admissions.
Segment	Documentation of one or more LOBs across **all levels**.	A focused slice of the EA.	Finance LOB segment covering goals → infrastructure.
Component	Changeable goal, process, system, or resource.	Can exist at any level within a LOB or segment.	Payroll system, Student portal, Data warehouse.
Vertical Component	Component that supports only **one LOB**.	Narrow, LOB-specific.	HR payroll system.
Crosscutting (Horizontal) Component	Component that supports **multiple LOBs**.	Enterprise-wide.	Email system, Intranet, Security tools.
Thread	Cross-cutting concern present in all levels.	Ensures consistency across the EA.	Security, Standards, Skills.
Artifact	Documentation describing a component.	Stored in EA repository.	Process maps, ER diagrams, system inventory.

Reference Architecture / Segment Architecture

- **A reference architecture** is the part of an EA that provides standards and documentation for a particular type of capability throughout the enterprise – such as mobile services or cloud computing.

- **A segment architecture** is somewhat similar, but usually focuses one or more particular business units or functions – such as the finance and accounting group, or how a financial ERP system and all of its modules are going to be implemented (general ledger, accounts payable, accounts receivable, payroll, benefits, etc.).



Fitting the Architecture Elements Together

- While the basic elements of EA analysis and design provide holistic and detailed descriptions of the current and future architecture in all areas of the underlying framework, it is important to also be able to articulate these relationships in discussions and presentations with executives, managers, support staff, and other EA stakeholders.
- Understanding and relating how the architecture fits together is essential to use the EA in planning and decision-making throughout the enterprise.
- This communication is supported through **two EA program resources**: the *EA Management Plan* and the *EA Repository*.
  - **EA Management Plan:** is a living document that is periodically updated so that it remains relevant as the ongoing primary reference for describing where the current and future architectures are at.
  - **The EA Repository:** is the on-line archive for EA information and the documentation artifacts that are described in the EA Management Plan.

online documentation of all required artifacts

Example of EA implementation plan

The following is an example of how to communicate about EA with stakeholders. In this example, some questions are presented about how to apply an EA framework to an enterprise, which subsequent classes answer.

- These are the types of questions that should be answered in the first few sessions of EA program and documentation meetings in order to promote an understanding of how the EA framework and documentation can reflect the enterprise.

- In the following example of how to talk about EA, the **five** levels and **three** vertical threads of the EA3 Cube framework are used for illustration.

Example of EA implementation plan 2

- Each area of the EA3 Framework represents a functional area of the enterprise. The EA3 Framework can be used in a top-down, bottom-up, or single-component manner.
- To begin to use the framework in a top down-manner, a series of questions at each level should be asked in order to determine how information about the enterprise will fit within that level of the framework.

Example of EA implementation plan 2

- The first questions to ask relate to the strategic **'Goals and Initiatives'** level of the framework.
- **The questions are:**
  - (1) For what purpose does the enterprise generally exist (usually expressed in the mission statement) and
  - (2) What kind of organization does the enterprise generally intend to be (often given in the vision statement)? What are the primary goals (strategic goals) of the enterprise? What then are the strategic initiatives (ongoing programs or new projects) that will enable the enterprise to achieve those goals?
  - Finally, for this level, when will the enterprise know that it has successfully reached these strategic goals or is making progress toward these goals (outcome measures)?

Example of EA implementation planning 3

- Second is the business **'Products and Services'** level of the framework, and it is important to first ask:
  - (1) What are the ongoing activity areas (lines of business) that the enterprise must engage in to support and enable the accomplishment of both strategic initiatives and normal 'maintenance'/ housekeeping' functions?
  - (2) What then are the specific activities in each line of business (business services)?
  - (3) What are the products that are delivered in each line of business?
  - (4) How do we measure the effectiveness and efficiency of the line of business processes (input/output measures) as well as their contribution to strategic goals (outcome measures)?
  - (5) Do any of these business services or manufacturing processes need to be reengineered/improved before they are made to be part of the future architecture?
  - (6) What are the workforce, standards, and security issues at this framework level?

Example of EA implementation planning 4

- Third is the **'Data and Information'** level of the framework. When the lines of business and specific business service/products have been identified, it is important to ask:
  - (1) What are the flows of information that will be required within and between activity areas in order to make them successful?
  - (2) How can these flows of information be harmonized, standardized, and protected to promote sharing that is efficient, accurate, and secure?
  - (3) How will the data underlying the information flows be formatted, generated, shared, and stored?
  - (4) How will the data become useable information and knowledge?

Example of EA implementation planning 4

- Fourth is the **'Systems and Applications'** level of the framework and it is important to ask:
  - (1) Which IT and other business systems and applications will be needed to generate, share, and store the data, information, and knowledge that the business services need?
  - (2) How can multiple types of IT systems, services, applications, databases, and web sites be made to work together where needed?
  - (3) How can configuration management help to create a cost-effective and operationally efficient 'Common Operating Environment' for systems and applications?
  - (4) What are the workforce, standards, and security issues at this level?

Example of EA implementation planning 5

ex: Bank - ATM integrated cloud-based mobile banking inf: data centers → encrypted communication

- Fifth is the **'Network and Infrastructure'** level of the framework and it is important to ask

  - (1) What types of voice, data, and video networks or computing clouds will be required to host the IT systems/applications and to transport associate, data, images, and conversations, as well as what type of infrastructure is needed to support the networks (e.g. buildings, server rooms, other equipment).

  - (2) How can these networks be integrated to create a cost-effective and operationally efficient hosting and transport environment?

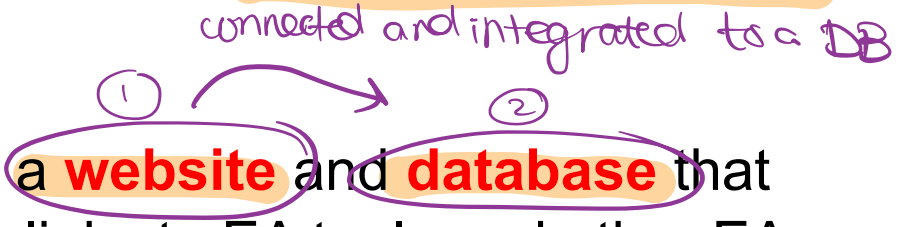
  - (3) Will these networks and clouds extend beyond the enterprise?

  - (4) What are the workforce, standards, and security issues at this level?

  - (5) What are the physical space and utility support requirements for these infrastructure resources?

The EA Repository

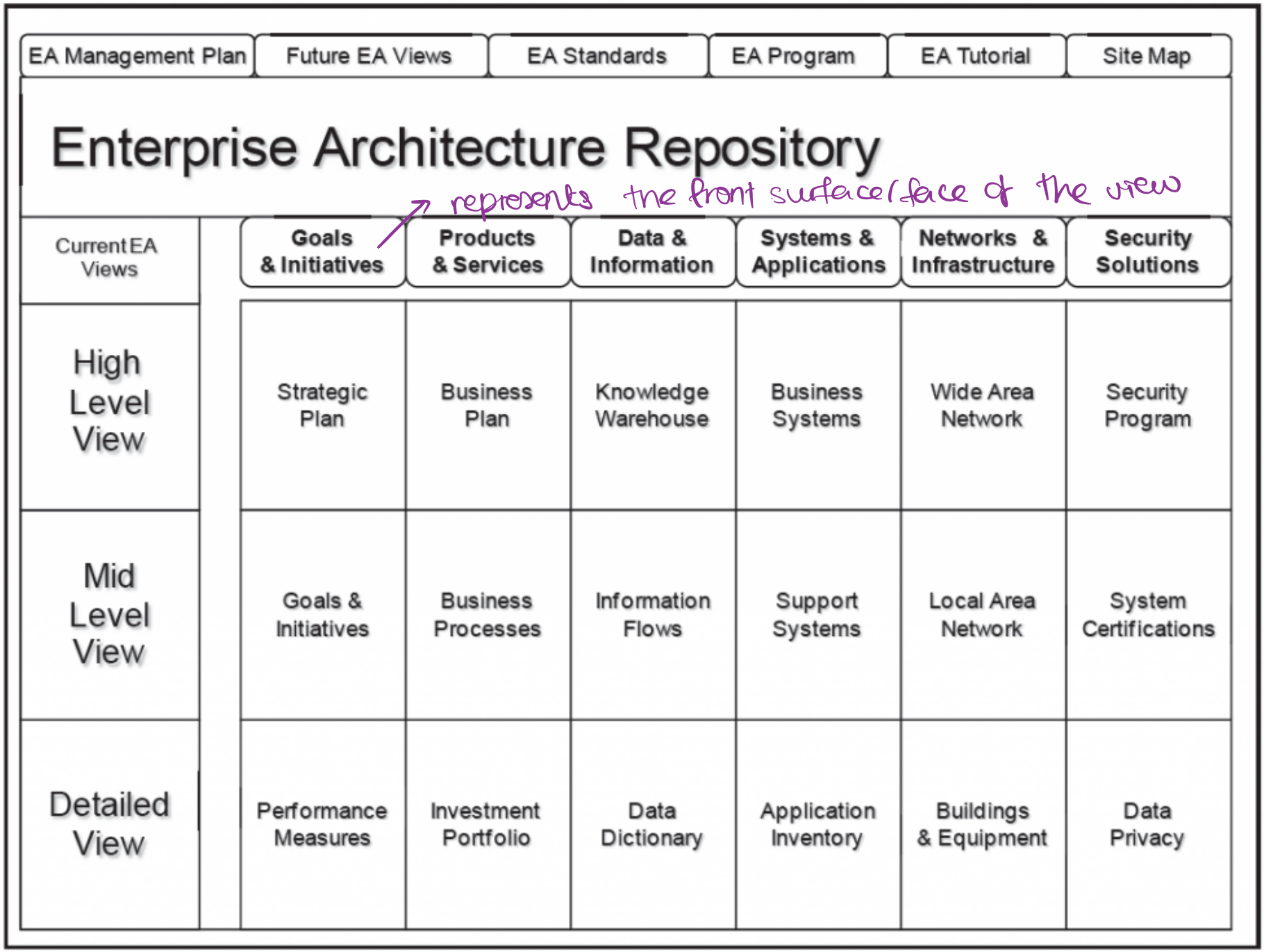
- Providing easy access to EA documentation is essential for use in planning and decision-making. This can be accomplished through the establishment of an on-line EA repository to archive the documentation of EA components in the various areas of the EA framework.



- The EA repository is essentially a **website** and **database** that stores information and provides links to EA tools and other EA program resources.

- Figure in next slide is an example of how an EA repository might be designed. This example is called Living Enterprise and is designed to support documentation that is organized through the use of the EA3 Cube Framework.

The EA Repository




The End